**In this question you must show detailed reasoning.**

**Solutions relying entirely on calculator technology are not acceptable.**

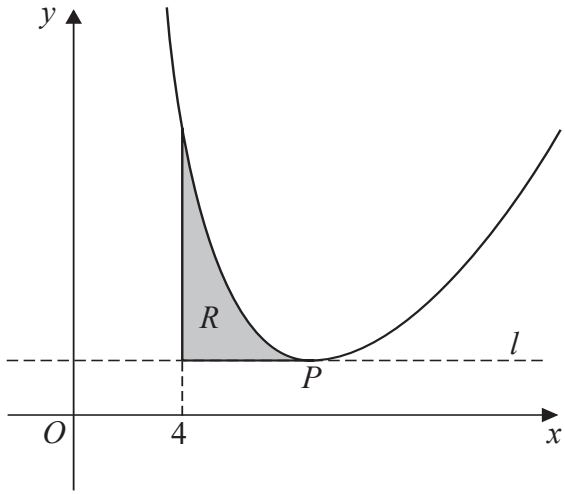



**Figure 2**

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \qquad x > 0$$

The point $P$ is the only stationary point on the curve.

(a) Use calculus to show that the $x$ coordinate of $P$ is 9

**(4)**

The line $l$ passes through the point $P$ and is parallel to the $x$-axis.

The region $R$, shown shaded in Figure 2, is bounded by the curve, the line $l$ and the line with equation $x = 4$

(b) Use algebraic integration to find the exact area of $R$.

**(5)**